Pregunta 22.11

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Series de tiempo

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Colab
https://colab.research.google.com/drive/1nhDuLDCbX33uq8RVJ_Iv9i5BZgk8PeI8?usp=sharing

Prompt
https://grok.com/chat/b7ab9482-3674-40f9-997a-2bd316674ce6

Github
https://github.com/A00573236/Pregunta-22.11

Interpretations

Stochastic Nature

The money supply series exhibits stochastic behavior, meaning its values are influenced by random shocks and historical patterns rather than deterministic rules. This is typical for macroeconomic variables like money supply, which are affected by unpredictable factors such as central bank policies, economic shocks, or changes in credit demand. The selected ARIMA(2,1,3) model captures this stochasticity through:

- Two autoregressive (AR) terms (p=2): The current money supply depends on its values from the two previous periods, reflecting persistence in the series.
- Three moving average (MA) terms (q=3): The series is also influenced by random shocks (errors) from the three prior periods, indicating that unexpected changes in money supply have lingering effects. The stochastic component suggests that forecasts are probabilistic, and external events (e.g., monetary policy shifts) could introduce volatility, as seen in the wide 95% confidence interval ([908.60, 1162.12] at period 30).

Stationarity

The money supply series is non-stationary, as confirmed by two unit root tests:

- ADF (Augmented Dickey-Fuller) Test:
  - Statistic: -0.4674
  - p-value: 0.8982 (>> 0.05)
  - Interpretation: The failure to reject the null hypothesis of a unit root indicates that the series is non-stationary, likely due to a trend or random walk behavior.
- KPSS Test:
  - Statistic: 3.6046
  - p-value: 0.0100 (< 0.05)

    - Interpretation: The rejection of the null hypothesis of stationarity further confirms that the series has non-stationary properties, such as a time-varying mean or trend. To achieve stationarity, the ARIMA model applied one differencing (d=1), which removed the non-stationary trend, allowing the model to focus on short-term fluctuations. This is consistent with many economic series, where differencing is often required to stabilize the mean.

Unit Root

The unit root detected by the ADF test (p-value = 0.8982) indicates that the money supply series has a stochastic trend, meaning its changes are persistent and do not revert to a fixed mean without differencing. This is a hallmark of non-stationary series and aligns with economic theory, as money supply often grows or contracts over time due to policy decisions or economic conditions. The unit root necessitated the differencing step in the ARIMA model, transforming the series into a stationary process for reliable modeling and forecasting.

Correlograms

While correlograms (ACF and PACF plots) were not explicitly generated in the output, the selection of the ARIMA(2,1,3) model implies that they were indirectly considered during model specification. In practice:

- The ACF (Autocorrelation Function) of the differenced series likely showed significant lags up to order 3, justifying the inclusion of three MA terms (q=3) to account for the correlation of past errors.
- The PACF (Partial Autocorrelation Function) likely indicated significant lags at orders 1 and 2, supporting the two AR terms (p=2) to capture the direct influence of prior values. These correlograms would have guided the model selection by identifying the appropriate orders for the AR and MA components, ensuring the model captures the series' autocorrelation structure effectively.

Cointegration

Cointegration was not tested in this analysis because it involves only a single time series (money supply). Cointegration is relevant when analyzing multiple non-stationary series to determine if they share a stable long-run equilibrium relationship. Since the analysis focused on one series, cointegration does not apply directly. However, in a broader context, the money supply could be tested for cointegration with related variables (e.g., inflation, interest rates, or GDP) to explore whether they move together over time despite being non-stationary. The absence of such tests limits insights into long-run relationships but does not affect the validity of the single-series ARIMA model.

Forecast and Economic Implications

The ARIMA(2,1,3) model, with an AIC of 2298.30, forecasts a downward trend in the money supply:

- Last observed value: 1093.46
- Average forecast value: 1065.22
- Forecasted change: -28.24 (a decline over 30 periods)
- Next 5 periods:
    - 489: 1089.50
    - 490: 1089.28
    - 491: 1088.75
    - 492: 1087.93
    - 493: 1086.85
- 95% CI at period 30: [908.60, 1162.12], indicating significant uncertainty in long-term forecasts.

This downward trend suggests a potential contractionary monetary environment, possibly due to tighter monetary policy, reduced credit creation, or lower economic activity. Economically, a declining money supply could lead to lower inflation or higher interest rates, but it may also signal risks of economic slowdown if not balanced by other factors.

Model Reliability and Warnings

The model faced convergence warnings and issues with non-stationary/invertible parameters, indicating challenges in fitting the ARIMA(2,1,3) model. These warnings suggest that the model's complexity (p=2, q=3) or the data's properties (e.g., volatility) may have caused numerical instability. However, the low AIC confirms that this model is the best among those tested, and the forecasts remain credible for short-term predictions. The wide confidence interval reflects the stochastic nature of the series and the uncertainty inherent in long-term economic forecasting.

Limitations and Next Steps

- Stochastic shocks: The model assumes that future shocks follow historical patterns, but unexpected policy changes or economic events could disrupt the forecast.
- Correlograms: Explicitly plotting ACF/PACF could refine model selection and confirm the appropriateness of p=2 and q=3.
- Cointegration: Testing cointegration with related series (e.g., inflation or GDP) could reveal long-run relationships, enhancing the analysis.
- Stationarity: Further transformations (e.g., logarithmic differencing) could address convergence issues and improve model stability.
- Short-term focus: Given the uncertainty in the 95% CI, the model is more reliable for short-term forecasts (5-10 periods) than the full 30-period horizon.

Summary of Steps Involved to Carry Out the Task

1. Load and clean the money supply data, visualizing for trends.
2. Confirm non-stationarity with ADF and KPSS tests, applying differencing (d=1).
3. Use ACF/PACF plots to identify p and q, testing models like ARIMA(p,1,q).

4. Select the best model (e.g., ARIMA(2,1,3)) based on AIC, addressing convergence warnings.
5. Validate the model with residual diagnostics (ACF, Ljung-Box test).
6. Generate and visualize forecasts with confidence intervals.
7. Optionally validate with out-of-sample testing.
8. Interpret the model economically and document findings.

Own interpretation

From my perspective, this analysis underscores the inherent complexity of economic series like the money supply, which are shaped not only by historical patterns but also by unpredictable events beyond the model's control. The choice of the ARIMA model seems suitable for capturing short-term dynamics, but the convergence warnings make me cautious about its robustness, particularly for longer horizons. The forecasted downward trend is intriguing; it could signal tighter monetary policies or weaker economic activity, with potential implications like deflationary pressure or rising interest rates. Yet, the wide confidence interval reminds me that economic forecasts are inherently uncertain, and external events could significantly alter the outlook.

I find the absence of cointegration analysis particularly noteworthy, as exploring long-term relationships with variables like inflation or GDP could provide richer context for the money supply's evolution. Additionally, the reliance on implied correlograms to justify the AR and MA terms suggests that explicit ACF/PACF plots could have strengthened model selection. Personally, this exercise highlights the need to blend statistical modeling with a deep understanding of economic context, as data alone doesn't tell the full story. To enhance the analysis, I'd suggest incorporating cointegration tests and exploring simpler models like ARIMA to address convergence issues, while prioritizing short-term forecasts where the model appears more reliable.